

OCEANEMBED

SIH 2026 — SIH26066

Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature

Research & Technical Team Dossier

"We are not selecting the most sophisticated model. We are selecting the simplest architecture that demonstrably solves the SIH problem better than its simpler alternatives, while remaining scientifically defensible and feasible for our team."

1. Executive Summary

OceanEmbed is our team's proposed framework for Smart India Hackathon (SIH) 2026 Problem Statement 26066. The objective is to reconstruct 3D subsurface ocean temperature profiles using only 2D surface observations collected by satellites.

This document serves as the single source of truth for our 6-member team. It establishes the scientific foundation, documents our experimental audits, evaluates candidate solutions, defines our minimum viable product (MVP), and outlines a rigorous research roadmap. We prioritize a **Baseline-First** methodology, meaning we will validate simple machine learning models before adopting complex deep learning architectures. Every component will be scientifically validated against independent ARGO float data.

2. CURRENT REALITY: Where we are right now

✓ PS understood ✓ GLORYS identified ✓ Initial dataset audited ✓ Initial dataset found unsuitable for 3D training (Dataset acquisition audit successfully identified an unsuitable subset before model training). ✓ Native depth structure investigated
Correct 3D GLORYS pilot pending (Blocked by authentication). **Seven surface datasets being investigated** **Baseline ML not yet trained** **ARGO validation not yet performed** **Final architecture not yet experimentally selected**

3. Problem Statement

[Official Source: SIH 26066] The ocean surface is continuously monitored by satellites, providing dense spatial and temporal data. However, the subsurface ocean—which holds the vast majority of the ocean's heat—remains hidden from space. Direct subsurface observations are limited to sparse measurements from ARGO floats, buoys, and ships. The problem is to develop a framework that maps observable surface variables to the hidden subsurface temperature structure.

4. Why This Problem Matters

Why care about subsurface ocean temperature?

- **Ocean Circulation:** Temperature gradients drive massive ocean currents.
- **Upper-Ocean Heat Content:** Critical for understanding climate variability.
- **Stratification:** Determines mixing and nutrient availability.
- **Air-Sea Interaction:** Fuels tropical cyclones and monsoons.
- **Marine Ecosystems & Fisheries:** Determines habitats and marine heatwaves.
- **Data Assimilation:** Improves operational weather forecasting.

Direct observational tools (ARGO, buoys, gliders, ships) cannot alone provide continuous basin-scale fields because they only sample specific, isolated points in a vast ocean.

5. OceanEmbed in 60 Seconds

We can observe the ocean surface relatively well from satellites, but we cannot continuously observe what is happening hundreds of metres below it. OceanEmbed attempts to learn the hidden relationship between the surface state and the subsurface temperature profile.

Pipeline Overview:

```
graph TD
  A[Surface Ocean Observations] --> B[Data Harmonization]
  B --> C[0.25° Daily Grid]
  C --> D[Satellite Representation / Embedding]
  D --> E[Deep Learning Reconstruction]
  E --> F[15-Depth Temperature Profile]
  F --> G[ARGO Independent Validation]
```

6. Exact SIH Requirements

[Official Source: SIH Specification]

Requirement	SIH Specification
Region	North Indian Ocean
Latitude	5°N–30°N
Longitude	45°E–105°E
Spatial output	0.25° × 0.25°
Temporal output	Daily
SST	Required
SSS	Required
SSH/SLA	Required
Surface U	Required
Surface V	Required
Wind U	Required
Wind V	Required
Target	Subsurface temperature
Depths	0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000 m
Training reference	GLORYS Global Ocean Reanalysis
Independent validation	Gridded ARGO / INCOIS LAS
Metrics	RMSE, correlation, Bias, etc.
Expected demonstration	Bay of Bengal / Arabian Sea

7. What Are We Actually Solving?

OceanEmbed is fundamentally a **high-dimensional spatiotemporal inverse problem**: infer a 15-level vertical ocean temperature profile at every 0.25° daily grid location from a multi-variable surface observation field.

It operates across three dimensions:

- **Horizontal**: 0.25° × 0.25° spatial context.

- **Vertical:** 15 target depths from surface to 1000 m.
- **Temporal:** Daily evolution.

The problem is NOT simply taking 7 numbers and doing a linear regression to 15 numbers. It is mapping a: Spatiotemporal surface field → Latent ocean state → Vertical temperature profile

8. Input Variables

The model receives seven surface variables. (Note: "Expected Relevance" below is a Research Hypothesis to be proven experimentally).

Variable	Physical information	Expected relevance
SST	Surface thermal state	High
SSS	Freshwater/haline structure (density driver)	High
SSH/SLA	Dynamic height / thermocline-related information	High
U/V current	Surface transport / mesoscale dynamics	High
Wind U/V	Surface forcing / Ekman pumping	Medium–High

9. Output Representation

The model predicts: T(0m), T(5m), T(10m), T(20m), T(30m), T(50m), T(75m), T(100m), T(125m), T(150m), T(200m), T(300m), T(500m), T(700m), T(1000m) .

The output is a continuous **vertical profile**, not 15 unrelated regression problems. This physical reality must influence our architecture design (e.g., treating it as a sequence or interconnected physical structure, not isolated outputs).

10. Data Architecture

```
graph TD
    A[Multi-Source Satellites] --> B[Quality Control & Temporal Harmonization]
    B --> C[Spatial Regridding 0.25°]
    C --> D[Missing Data Masking]
    D --> E[7-Channel Input Tensor]
    E --> F[OceanEmbed Model]
    F --> G[15-Depth Profile]
```

11. GLORYS

[Official Source: cmems_mod_glo_phy_my_0.083deg_P1D-m] GLORYS is the Global Ocean Reanalysis produced by Mercator Ocean. It assimilates satellite and in-situ data into physical simulations (NEMO) to produce daily, native 1/12° resolution, 3D ocean states.

Important: GLORYS is our **training reference / reanalysis target**, NOT "ground truth." Reanalysis itself is an assimilated model output. True independent validation requires ARGO.

12. ARGO

Independent validation against ARGO floats is required because reanalysis is not equivalent to direct observational ground truth. Validation will require careful spatial colocation, temporal matching, depth interpolation, and quality control.

13. Current Data Feasibility

[Experimentally Verified]

- **Native Depths:** Verified 36 native GLORYS depth levels covering 0.49m to 1062.44m needed for the SIH 1000m target.
- **Initial Audit:** A downloaded GLORYS subset successfully parsed but was found to be a 2D surface slice. This audit successfully prevented us from building a pipeline on invalid data.
- **Blocker:** Corrected 3D download is currently blocked by Copernicus authentication.

14. The Obvious Solutions

Before designing complex architectures, we evaluate standard approaches.

Approach 1: Point-wise MLP

- *Potential Problem:* Each grid cell is treated independently. It misses eddies, neighboring thermal structures, and spatial gradients.
- *Status:* Keep as Level 1 Baseline.

Approach 2: Random Forest / XGBoost

- *Potential Problem:* Does not naturally model spatial fields or learn spatial embeddings.
- *Status:* Keep as benchmark if needed, but not final architecture.

Approach 3: Plain CNN

- *Important:* A CNN is likely one of the strongest baselines because the problem is highly spatial. If a CNN performs almost as well as a sophisticated model, additional complexity is not justified.
- *Status:* Level 3 Baseline.

Approach 4: U-Net

- *Advantage:* Proven for dense spatial reconstruction, encoder-decoder structure.
- *Weakness:* Not inherently temporal.
- *Status:* Strong Candidate.

Approach 5: ConvLSTM / Temporal Models

- *Advantage:* Explores daily temporal evolution.
- *Weakness:* Training cost, memory limits. We must test whether temporal history *materially* improves performance over spatial context alone.
- *Status:* Candidate for advanced integration.

Approach 6: Pure ViT / Transformer

- *Advantage:* Global spatial relationships.
- *Weakness:* Computational cost, high data requirements, potentially unnecessary for a regional 0.25° grid.
- *Status:* Candidate, but CNN/attention hybrids might be more feasible.

Approach 7: Pure Autoencoder

- *Advantage:* Compresses data.
- *Weakness:* Compression ≠ prediction of subsurface temperature.
- *Status:* Rejected as standalone, but latent encoder concepts will be used.

Approach 8: GNN (Graph Neural Networks)

- *Advantage:* Irregular spatial relationships.
- *Weakness:* The 0.25° grid is strictly regular. GNNs add massive computational complexity without obvious benefit over CNNs for regular grids.
- *Status:* Rejected unless regular grid assumptions fail.

15. Solution Elimination Matrix

Approach	Strength	Fatal/Major Weakness	Feasibility	Novelty	Verdict
Climatology	Easiest baseline	Ignores actual daily dynamics	High	None	Baseline 0
Point-wise MLP	Fast, easy to implement	Ignores spatial context/eddies	High	None	Baseline 1
Random Forest	Strong nonlinear regressor	Fails at spatial field tasks	High	None	Reject as final
Plain CNN	Captures spatial context	Treats depths independently	High	Low	Baseline 3
U-Net	Multi-scale spatial features	Standard versions ignore time	High	Medium	Strong Candidate
ConvLSTM	Captures spatial + temporal	Memory intensive, slow to train	Medium	Low	Candidate
Pure ViT	Long-range global context	Data hungry, hard to optimize	Low-Med	Medium	Hybridize instead
GNN	Irregular grids	Unnecessary for 0.25° regular grid	Low	High	Reject

16. Existing Research Landscape

[Literature Review Requirement] We must acknowledge peer-reviewed literature. Existing research has already demonstrated:

- ConvLSTM-based global subsurface reconstruction.
- CNN/LightGBM reconstruction to 1000m at 0.25° resolution.
- Convformer and Graph-attention approaches.
- Attention-enhanced U-Net approaches.
- Uncertainty-aware AI reconstruction.

17. What Has Already Been Done?

Current literature proves that converting satellite data to subsurface temperature is possible using CNNs, ConvLSTMs, and Transformers. **We cannot claim that simply using a CNN or ConvLSTM for this task is a novel contribution.**

18. Research Gap (OceanEmbed Opportunity)

Candidate Research Gaps (Hypotheses to be validated):

- **Multi-variable fusion:** Do existing models optimally fuse all 7 SIH variables?
- **Missing-data robustness:** Can the model maintain accuracy under heavy cloud cover (missing SST)?
- **Uncertainty:** Can the model output "24.7°C ± 0.8°C" rather than false confidence?
- **Depth-aware decoding:** Do models learn that the 15 depths are physically connected, rather than independent targets?
- **ARGO-first validation:** Many models only validate against reanalysis. Independent ARGO validation remains a critical gap.

19. Candidate Architecture Comparison

Instead of guessing, we will experimentally compare architectures based on their ability to minimize error against ARGO, handle missing data, and run efficiently on constrained hardware.

20. Feasibility Scoring

We propose the following weighted scoring framework for final architecture selection:

Metric	Proposed Weight	Goal
Scientific Suitability	20%	Physical/Oceanographic sense
Accuracy Potential	15%	Minimizes RMSE
Feasibility	15%	Trainable by 6 students
Spatial/Temporal Cap.	10%	Uses the 3D nature of data
Data Efficiency	10%	Trains on limited regional data
Novelty Potential	10%	SIH differentiator
Impl. Complexity	5%	Code maintainability
Interpretability	5%	Explainable to judges
Validation Compat.	5%	Maps well to ARGO
SIH Demo Suitability	5%	Runs fast for UI inference

21. Experimental Ladder

We will climb this ladder, stopping when added complexity no longer yields significant ARGO improvement.

- **LEVEL 0:** Climatology / Persistence
- **LEVEL 1:** Point-wise MLP
- **LEVEL 2:** XGBoost / LightGBM
- **LEVEL 3:** Spatial CNN / U-Net
- **LEVEL 4:** Temporal CNN / ConvLSTM
- **LEVEL 5:** Attention-enhanced spatial model
- **LEVEL 6:** Final OceanEmbed Architecture

22. PROPOSED OCEANEMBED SOLUTION

We are not trying to invent an entirely new neural network.

Instead: **We are designing a scientifically validated pipeline that learns a compact representation of the multi-variable surface ocean state, captures spatial/temporal context, and decodes that representation into a physically coherent 15-depth temperature profile.**

Our conceptual framework merges spatial encoding with depth-aware decoding and uncertainty estimation, ensuring it remains robust even when satellite data is missing.

23. Architecture Deep Dive

[Proposed Conceptual Architecture — To be experimentally validated]

```
graph TD
    A[7 Surface Variables] --> B[Quality / Missing Mask]
    B --> C[Spatial Encoder CNN / U-Net blocks]
    C --> D[Latent Ocean State]
    D --> E[Spatial Context]
    D --> F[Temporal Context]
```

```
E --> G[Attention / Fusion]
F --> G
G --> H[Depth-aware Decoder]
H --> I[0m]
H --> J[100m]
H --> K[1000m]
I --> L[15-Depth Profile]
J --> L
K --> L
L --> M[Uncertainty Estimate]
```

24. Depth-Aware Reconstruction

[Research Hypothesis] Instead of treating the 15 outputs as unrelated neurons, we investigate a shared latent ocean representation combined with depth information. This allows the model to understand that the 100m and 125m predictions are physically adjacent points on the same underlying vertical ocean structure.

25. Missing Data Strategy

[Research Hypothesis] Satellite data often has gaps (clouds masking SST, orbital gaps in altimetry). We will compare:

- **Model A:** Requires all 7 inputs (Baseline).
- **Model B:** Uses an explicit availability mask.
- **Model C:** Uses learned missing-data handling / spatial interpolation. If B or C performs competitively under realistic missingness, this becomes a major SIH differentiator.

26. Temporal Context

[Research Hypothesis] Does knowing yesterday's ocean state improve today's 1000m depth prediction? We will experimentally verify if temporal history (e.g., via ConvLSTM or temporal sliding windows) materially improves performance over a purely spatial snapshot.

27. Uncertainty Strategy

[Research Hypothesis] Recent research demonstrates uncertainty-aware ocean temperature reconstruction. We will investigate:

- MC Dropout
- Deep Ensembles We must determine if uncertainty is genuinely useful and feasible within SIH constraints before finalizing it.

28. Physics-Aware Strategy

[Research Hypothesis] We will **NOT** blindly add a physics loss. A simplistic constraint assuming temperature must always decrease with depth is scientifically invalid in regions with barrier layers (like the Bay of Bengal). We will experimentally evaluate:

- Vertical smoothness constraints.
- Climatological consistency.
- No physics loss (letting data drive the structure entirely). If a physics loss does not improve ARGO performance, we will drop it.

29. Training Strategy

- **Optimizer:** To be evaluated (AdamW likely).
- **Loss:** MSE or RMSE + potential uncertainty/physics regularizers (if proven).
- **Hardware:** Consumer/Cloud GPUs available to students.

30. Validation Strategy

Validation must be entirely independent.

- **GLORYS:** Used to train the network.
- **ARGO:** Used exclusively for independent, observational holdout validation. We will not expect exact coordinate matching; we will colocate floats within reasonable spatial/temporal tolerances.

31. Leakage Prevention

[Proposed Split] Data leakage destroys scientific credibility. We must prevent temporal, spatial, and normalization leakage.

- **Training:** 2012–2020
- **Validation:** 2021
- **Test:** 2022 *All normalization statistics MUST be calculated exclusively on the training set.*

32. MVP (Minimum Viable Product)

The minimum scientifically credible system:

1. Real surface data
2. 0.25° grid harmonization
3. Real GLORYS target
4. Point-wise MLP baseline
5. One spatial deep-learning model (CNN)
6. 15-depth reconstruction
7. ARGO validation

33. Advanced Version

Only after the MVP works will we attempt to add:

- Temporal context
- Attention mechanisms
- Missing-data robustness
- Uncertainty
- Advanced embedding ablations

34. Go / No-Go Gates

1. **Gate 1:** Can we obtain all seven inputs?
2. **Gate 2:** Can we build a real 0.25° daily dataset?
3. **Gate 3:** Can a simple baseline reconstruct the profile?
4. **Gate 4:** Does spatial context improve it?
5. **Gate 5:** Does temporal context improve it?
6. **Gate 6:** Does the embedding improve performance?
7. **Gate 7:** Does the model improve against ARGO?
8. **Gate 8:** Is the improvement worth the additional complexity? (If no, keep the simpler model).

35. Six-Member Execution Plan

Member	Track	Responsibilities
Member 1	Data Acquisition	GLORYS + satellite datasets access and download.
Member 2	Data Engineering	Regridding, temporal alignment, QC, missing values.
Member 3	ML Baselines	MLP / CNN / U-Net baseline training and metrics.

Member 4	Embedding Research	CNN / ViT / Autoencoder / Temporal experiments.
Member 5	ARGO Validation	ARGO acquisition, matching, leakage audit, metrics.
Member 6	Product & Viz	Dashboard, inference interface, SIH presentation.

36. Timeline

- **Phase 1 (Data feasibility):** Verify datasets.
- **Phase 2 (Real pilot):** Build end-to-end pipeline (7-day test).
- **Phase 3 (Baselines):** Train MVP models.
- **Phase 4 (Experiments):** Test candidate representations.
- **Phase 5 (Validation):** Independent ARGO testing.
- **Phase 6 (Refinement):** Ablation and robustness tests.
- **Phase 7 (Final Model):** Final architecture selection.
- **Phase 8 (UI):** Connect model to SIH Demo Dashboard.
- **Phase 9 (Defense):** Presentation preparation.

37. Risks

- **Data feasibility:** Securing reliable daily feeds for all 7 variables.
- **Compute constraints:** High-dimensional models exceeding student hardware.
- **Validation size:** Finding sufficient ARGO matches in the strict 2022 temporal holdout window.

38. Judge Questions

- **What is the simplest viable solution?** An MLP regression on point-wise surface data.
- **What obvious approaches are rejected?** Global GNNs, pure un-hybridized ViTs.
- **Why are they rejected?** Unnecessary complexity for a regular 0.25° grid, demanding unfeasible compute for 6 students.
- **What is our strongest baseline?** A spatial CNN or standard U-Net.
- **What is potentially novel?** Missing-data masking, depth-aware decoders, and uncertainty bounds (if proven experimentally).
- **Can this operate daily?** Inference latency will be measured after the final model and hardware environment are fixed, but CNN inference is typically highly efficient.

39. Current Decision Log

ID	Decision	Status	Reason
D-001	Target 7 native 0.25° sources	Approved	Minimizes interpolation leakage.
D-002	Run audit scripts locally	Approved	Remote env constraints on large NetCDFs.
D-003	7-day GLORYS pilot (3D depths)	Approved	Validates pipeline safely before massive downloads.

40. References

1. [Official Source] Smart India Hackathon 2026 Problem Statement 26066.
2. [Official Source] Copernicus Marine Service. *Global Ocean Physics Multiyear Product*. Dataset: `cmems_mod_glo_phy_my_0.083deg_P1D-m`.
3. [Official Source] GLORYS12V1 Reference. DOI: <https://doi.org/10.48670/moi-00021>
4. [Official Source] ARGO Data Management. Gridded ARGO / INCOIS LAS. *(Additional peer-reviewed literature citations pending formal literature review completion).*